SEPARATING A CLOSED SUBSET FROM A POINT AFTER FINITE PRESENTATION BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4554-4626, following the statement at 4528-4552.

Editorial note. No load-bearing omissions; the closing paragraph of the statement (V, Z' affine; V an affine open when Z is a scheme) is handled by the reductions in the proof.

Allowed standard prerequisites (editorial).

- Scheme-case separation lemma. Limits, Lemma limits-lemma-separate (cross-chapter, cited at ~4581). Produces V, Z', T' for W -> Y and the closed subset T_W.
- Points of cartesian squares under surjective etale maps. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian (cross-chapter, cited at ~4611-4613). a_1(T_1) = q^{-1}(a_2(T_2)).
- Open morphisms and quotient topology. Topology, Lemma topology-lemma-open-morphism-quotient-topology (cross-chapter, cited at ~4614-4615). q^{-1}(closure) equals closure of q^{-1} for the surjective open q.

Source reference. Original reference 0CM8: focused-seeds/geometry/S9-spaces-limits-separate/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-limits-separate/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02YB: focused-seeds/geometry/S9-spaces-limits-separate/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Z \to Y$ be morphisms of algebraic spaces over S. Let $z \in |Z|$ and let $T \subset |X \times_Y Z|$ be a closed subset with*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$z \notin \mathrm{Im}(T \to |Z|)$. If f is quasi-compact, then there exists an étale neighbourhood $(V, v) \to (Z, z)$, a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow[a]{} & Z' \\ \downarrow & & \downarrow b \\ Z & \xrightarrow{g} & Y, \end{array}$$

and a closed subset $T' \subset |X \times_Y Z'|$ such that

1. *the morphism $b : Z' \to Y$ is locally of finite presentation,*
2. *with $z' = a(v)$ we have $z' \notin \mathrm{Im}(T' \to |Z'|)$, and*
3. *the inverse image of T in $|X \times_Y V|$ maps into T' via $|X \times_Y V| \to |X \times_Y Z'|$.*

Moreover, we may assume V and Z' are affine schemes and if Z is a scheme we may assume V is an affine open neighbourhood of z.

Proof. We will deduce this from the corresponding result for morphisms of schemes. Let $y \in |Y|$ be the image of z. First we choose an affine étale neighbourhood $(U, u) \to (Y, y)$ and then we choose an affine étale neighbourhood $(V, v) \to (Z, z)$ such that the morphism $V \to Y$ factors through U. Then we may replace

1. $X \to Y$ by $X \times_Y U \to U$,
2. $Z \to Y$ by $V \to U$,
3. z by v, and
4. T by its inverse image in $|(X \times_Y U) \times_U V| = |X \times_Y V|$.

In fact, below we will show that after replacing V by an affine open neighbourhood of v there will be a morphism $a : V \to Z'$ for some $Z' \to U$ of finite presentation and a closed subset T' of $|(X \times_Y U) \times_U Z'| = |X \times_Y Z'|$ such that T maps into T' and $a(v) \notin \mathrm{Im}(T' \to |Z'|)$. Thus we may and do assume that Z and Y are affine schemes with the proviso that we need to find a solution where V is an open neighbourhood of z.

Since f is quasi-compact and Y is affine, the algebraic space X is quasi-compact. Choose an affine scheme W and a surjective étale morphism $W \to X$. Let $T_W \subset |W \times_Y Z|$ be the inverse image of T. Then z is not in the image of T_W. By the schemes case (Limits, Lemma 0CM8) we can find an open neighbourhood $V \subset Z$ of z a commutative diagram of schemes

$$\begin{array}{ccc} V & \xrightarrow[a]{} & Z' \\ \downarrow & & \downarrow b \\ Z & \xrightarrow{g} & Y, \end{array}$$

and a closed subset $T' \subset |W \times_Y Z'|$ such that

1. the morphism $b : Z' \to Y$ is locally of finite presentation,
2. with $z' = a(z)$ we have $z' \notin \mathrm{Im}(T' \to Z')$, and
3. $T_1 = T_W \cap |W \times_Y V|$ maps into T' via $|W \times_Y V| \to |W \times_Y Z'|$.

The commutative diagram

$$\begin{array}{ccccc} W \times_Y Z & \longleftarrow & W \times_Y V & \xrightarrow[a_1]{} & W \times_Y Z' \\ \downarrow & & \downarrow c & & \downarrow q \\ X \times_Y Z & \longleftarrow & X \times_Y V & \xrightarrow{a_2} & X \times_Y Z' \end{array}$$

has cartesian squares and the vertical maps are, surjective, étale and a fortiori open. Looking at the left hand square we see that $T_1 = T_W \cap |W \times_Y V|$ is the inverse image of $T_2 = T \cap |X \times_Y V|$ by c. By Properties of Spaces, Lemma 03H4 we get $a_1(T_1) = q^{-1}(a_2(T_2))$. By Topology, Lemma 02YB we get

$$q^{-1}\left(\overline{a_2(T_2)}\right) = \overline{q^{-1}(a_2(T_2))} = \overline{a_1(T_1)} \subset T'$$

As q is surjective the image of $\overline{a_2(T_2)} \to |Z'|$ does not contain z' since the same is true for T'. Thus we can take the diagram with Z', V, a, b above and the closed subset $\overline{a_2(T_2)} \subset |X \times_Y Z'|$ as a solution to the problem posed by the lemma. $\square$